

Raman Ebrahimi

+1-614-390-4131 | raman@ucsd.edu | [My webpage](#) | [LinkedIn](#) | [GitHub](#)

San Diego, CA, United States

SUMMARY

Quantitative Analyst with a strong mathematical foundation and expertise in AI. I possess strong analytical and programming skills, with experience in optimization, probabilistic modeling, and data analysis. Passionate about leveraging computational techniques, optimization, and statistical analysis, I aim to uncover insights, drive impactful solutions, and contribute to innovative projects by solving complex real-world problems with cutting-edge quantitative methods.

SKILLS

- **Programming Languages:** Python, SQL, JavaScript
- **Other Tools & Technologies:** Tableau, Gurobi, PowerBI, AnyLogic, NetLogo, $\LaTeX$
- **Machine Learning & NLP:** pandas, scikit-learn, sentiment analysis, intent classification, topic modeling, embeddings
- **Research Skills:** Mathematical modeling, Game Theory, Network Economics, Critical thinking, Problem-solving

EXPERIENCE

- **Data Solutions Engineer - Full-stack AI Engineering** April 2025 - present
HCM Tradeseal  Remote, USA
 - Engineered a Slack-based AI support agent that answers Professional Services questions from live Zendesk tickets using 3 read-only tools and 5 interchangeable LLM backends; implemented a 3-layer least-privilege architecture and validated it with 13/13 offline checks.
 - Automated a 4-stage customer sandbox workflow in Python and Playwright, from multi-email intake and data conversion through Auth0 onboarding and Valet import, supporting 7 import types and completing a verified end-to-end run in approximately 13 minutes; reduced rejected-upload diagnosis from a 15-minute timeout to seconds.
 - Architected WageFinder's end-to-end Python/FastAPI data platform, producing a 5.93M-record historical prevailing-wage corpus and expanding ingestion to 55 federal jurisdictions plus 7 state/local programs; authored 202 of 205 main-branch commits (98.5%) across scraping, parsing, APIs, data quality, security, and CI/CD.
 - Developed 5 estimator-facing products, wage forecasting, payroll compliance, scope classification, bid labor-cost estimation, and labor-market intelligence, plus an O*NET crosswalk covering 98.2% of wage rows; achieved 2.32% held-out MAPE, a 27% improvement over the 3.19% flat-escalation baseline.
 - Shipped 9 Valet tickets to master in 30 days across Vue 3/TypeScript, Express, and MongoDB, including server-side pagination and lazy loading for 2 high-volume interfaces and a full-stack admin group-deletion workflow; delivered tested changes across 47 files with integration, unit, and E2E coverage.
- **Machine Learning Engineer / Quantitative Researcher** May 2025 - June 2026
MarketCrunch AI  Remote, USA
 - Formulated rigorous statistical and mathematical frameworks to enhance predictive pricing models across a universe of 2,100+ equities and macroeconomic indicators.
 - Applied Explainable AI (XAI) techniques, specifically implementing SHAP values on our model alongside exploratory data analysis, to uncover hidden market signals, driving a consistent 3% increase in absolute prediction accuracy and a 6% boost in directional accuracy.
 - Redesigned the production prediction ensemble: implemented and benchmarked three candidate architectures (a LightGBM meta-learner over six base models, a two-stage direction-classifier routing to conditional magnitude models, and separately trained direction/magnitude specialists combined by soft voting) in a large-scale, date-aligned walk-forward backtest across the full ticker universe, parallelized on cloud VMs with per-ticker process isolation for reproducible, memory-safe runs.
 - Architected the backend for MarketPulse, a semi-live market signal feed: a FastAPI service generating ranked long/short signals every 15 minutes on top of a TensorFlow/scikit-learn prediction service containerized on Google Cloud Run (separate CPU and GPU deployments); restructured the prediction pipeline, reducing individual prediction latency by 35%.
 - Built and rigorously backtested an end-to-end automated trading system on the Alpaca API - signal generation, fundamental screening, convex portfolio optimization (CVXPY) under risk constraints, and automated order execution - supporting daily, intraday, simulation, and historical backtest modes.
- **Graduate Student Researcher** Sep 2022 - present
UC San Diego San Diego, USA
 - Formulated mathematical optimization models leveraging spectral graph theory and P-matrix properties to prove the existence, uniqueness, and stability of Nash equilibria across multiplex, multilayer, and K-hop network games.

- Pioneered the “Connected Minds” framework mathematically establishing how network opacity and homophily distort belief distributions and induce systematic cognitive biases.
- Developed machine learning frameworks and Python simulations to model strategic classification under bounded rationality, quantifying how human misperceptions of algorithmic feature weights impact decision systems.
- Validated theoretical insights through large-scale interactive user studies and evaluated Large Language Models (LLMs) as zero-shot semantic navigators to benchmark goal-directed search on complex information networks.

EDUCATION

- **University of California, San Diego** Sep 2023 - Mar 2028 (expected)
Ph.D. in Machine Learning and Data Science (ECE department) - *Advisor: Massimo Franceschetti* San Diego, USA
- **University of California, San Diego** Sep 2023 - Mar 2025
M.Sc. in Machine Learning and Data Science (ECE department) San Diego, USA
 - **Thesis:** From Games to Predictions: Strategic Behavior in Networks and Classification
- **Sharif University of Technology** Sep 2017 - Aug 2022
B.Sc. in Industrial Engineering (3.8/4.0) and B.Sc. in Physics (3.9/4.0) Tehran, Iran

PROJECTS

- **Self-Hosted Agentic Research Pipeline on a remote GPU Workstation** 2026
Tools: Python, LangGraph, FastAPI, Ollama, SQLite, Tailscale
 - Built an autonomous multi-agent research pipeline that runs open-weight LLMs on a campus GPU workstation via Ollama and is operated remotely from a laptop over Tailscale; nine playbooks span literature review, ideation, model refinement, proof attempts, simulation experiments, draft updates, and adversarial verification.
 - Orchestrated eight specialized agents with LangGraph state machines using Mixture-of-Agents ensembling: every step queries three local models (Gemma 26B, DeepSeek-R1 32B, Qwen 27B) and merges their outputs with a cloud synthesizer model.
 - Shipped a token-authenticated FastAPI server and browser dashboard with live log streaming (SSE), pause/resume, human-in-the-loop approval checkpoints, and SQLite checkpointing for crash-resumable runs; all artifacts are committed append-only to a git-backed research knowledge base.
- **Agentic-Research: Multi-Agent Research Plugin for Claude Code** 2026
Tools: Claude Code plugin system, subagents, skills, Git
 - Authored and published an open-source Claude Code plugin (distributed through a personal plugin marketplace) that turns a research workspace into a repeatable pipeline: eight specialized agents, nine slash-command playbooks, and eighteen skills covering proof writing, conjecture testing, adversarial proof review, citation management, and venue targeting.
 - Enforced epistemic rigor via a claim-lifecycle vocabulary and an evidence contract that blocks unproven “theorem” language, with human checkpoints built into the end-to-end pipeline; the local GPU pipeline above is a local-LLM port of this plugin.
- **Antipode ** 2026
Tools: Next.js (App Router), React, CSS
 - Built a 1-on-1, text-only debate platform that anonymously pairs users with opposing views: AI or human opponents, pre-chat icebreakers, shareable challenge links for social media, synchronized debate timers, and a spectator mode for reading past high-quality debates.
 - Designed incentive mechanics that reward persuasion over “winning”: a change-my-view concession metric and post-chat quality ratings that keep the ecosystem constructive without heavy moderation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [S.5] **Raman Ebrahimi**, Sepehr Ilami, Babak Heydari, Isabel Trevino, Massimo Franceschetti. **Visibility Design and Strategic Reasoning**. Manuscript submitted for publication in *Conference on Web and Internet Economics (WINE 2026)*.
- [S.4] **Raman Ebrahimi**. **Strategic Classification Has a Missing Lever: Audit Risk**. Manuscript submitted for publication in *Conference on Web and Internet Economics (WINE 2026)*.
- [S.3] **Raman Ebrahimi** et al. **Connected Minds: Network Transparency as a Design Lever for Boundedly Rational Multi-Agent Populations**. Manuscript submitted for publication in *Advances in Neural Information Processing Systems (NeurIPS 2026)*.
- [S.2] **Raman Ebrahimi** et al. **Regenerative Attention Markets: Surplus Cliffs and Equity Collapse under Engagement-Driven Competition**. Manuscript submitted for publication in *Advances in Neural Information Processing Systems (NeurIPS 2026)*.
- [C.5] **Raman Ebrahimi**, Sean Fuhrman, Kendrick Nguyen, Harini Gurusankar, Massimo Franceschetti. **Textual understanding boost in the WikiRace**. *The 14th International Conference on Complex Networks and their Applications*.

- [C.4] **Raman Ebrahimi**, Kristen Vaccaro, and Parinaz Naghizadeh. **The double-edged sword of behavioral responses in strategic classification: Theory and user studies**. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT 2025)*.
- [J.1] **R. Ebrahimi** and P. Naghizadeh, **United We Fall: On the Nash Equilibria of Multiplex and Multilayer Network Games**,. In *IEEE Transactions on Control of Network Systems*.
- [C.3] **Raman Ebrahimi**, Kristen Vaccaro, and Parinaz Naghizadeh. **The Double-Edged Sword of Behavioral Responses in Strategic Classification**. (*Spotlight talk*) *NeurIPS 2024 Workshop on Behavioral Machine Learning*.
- [C.2] **Ebrahimi, R.**, Naghizadeh, P. (2025). **Extended Horizons: Multi-hop Awareness in Network Games**. In: Sinha, A., Fu, J., Zhu, Q., Zhang, T. (eds) *Decision and Game Theory for Security. GameSec 2024*. Lecture Notes in Computer Science, vol 14908. Springer, Cham.
- [C.1] **R. Ebrahimi** and P. Naghizadeh, **United We Fall: On the Nash Equilibria of Multiplex Network Games**., 2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton), Monticello, IL, USA, 2023, pp. 1-8.

HONORS AND AWARDS

- **Awarded a \$40,000 scholarship for Master of Quantitative Finance program**
University of California, San Diego
- **Reviewer for American Control Conference, Automatica, and TRB**

April 2025

REFERENCES

Massimo Franceschetti

Professor, ECE, UCSD

Email: mfranceschetti@ucsd.edu

Isabel Trevino

Associate Professor, Economics, UCSD

Email: itrevino@ucsd.edu

Berk Ustun

Assistant Professor, Data Science, UCSD

Email: berk@ucsd.edu

Kristen Vaccaro

Assistant Professor, CSE, UCSD

Email: kv@ucsd.edu